Project Proposal NC

NC team 28

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Todo list

1 Context

Write a 2-page project proposal, in pdf, with the following elements: – A (tentative) title for your project – A problem statement with the background, relevance, and a concrete description of the problem you are solving (use exercise B.1 as input) – A section "related work" describing the relation of your project to other work in this area (use exercise B.2 as input) – A section "Approach" describing which methods you will use, why they are appropriate, which resources you need, and where/how you will obtain those (use exercise B.3) – A section "Planning" where you describe which milestones you want to finish when (use exercise B.4). Note that many of these elements will likely end up in your final report as well, so it is worth investing some time in this! Your final proposal should be concrete enough that it allows you to get started on the project. Hand in your proposal on Brightspace per the deadline listed there.

improved first pitch draft:

Neural Cellular Automata (NCAs) are small shared-weight networks that grow images from a single seed pixel and heal themselves after damage, and are often presented as very robust. Randazzo et al. (2021) and Cavuoti et al. (2022) showed you can still reprogram a trained Growing NCA,for instance making the target lizard lose its tail or turn red, using white-box, gradient-based attackers that have full access to the network's weights and backpropagate through its rollout.

Every published NCA attack uses gradients. Every published NCA robustness claim is correspondingly based on a stronger and less "natural" threat model than what an adversary in, say, a biological or bioengineering setting would plausibly have. We ask whether a query-only, black-box attacker using evolutionary search can reproduce the same targeted reprogramming attacks, and how close it gets to the gradient-based ceiling.

Concretely, we do not modify the NCA's weights and we do not perturb pixels directly. Following Randazzo et al., the attack surface is a 16×16 state-perturbation matrix, applied identically to every cell's 16-dimensional state vector (4 visible RGBA channels plus 12 hidden communication channels) at every timestep of the simulation. A 64×64 grid therefore has roughly 65,000 state values being transformed by the same 256 attack parameters each step. This compact, attack surface is exactly the type where CMA-ES could perform well (which doesn't perform well over 1000 parameters).

Our attacker sees only the visible RGB output at the end of the rollout, but never the weights, never the hidden channels, never any gradient information. Fitness is the L2 distance between that final image and a specific target image (e.g. a tailless lizard), so the attacker has to steer the NCA toward a chosen broken configuration. This matches Randazzo et al.'s targeted setup and makes the comparison of optimizers possible and also compare with random noise.

If evolutionary search can meaningfully reprogram the NCA under this restricted threat model, the robustness of Growing NCAs could be weaker than the literature implies. If it can't, the gap between white-box and black-box attacks on NCAs is a nice safety margin and shows they are robust against more 'realistic, or natural' attacks.

Things we can/should do if comparing attack types/optimizers isn't enough

- Untargeted ("destroy only") attack as a second condition: $fitness$ = maximize L2 from clean target, should be easy
- Adversarial training / arms race — take the attacks CMA-ES finds, add them to the damage distribution during NCA training. Does the NCA become robust to evolved attacks? Much bigger scope, only if we really have time
- Vary the attacker's observation window for CMA-ES: RBG only, RGBA, all 16 channels including hidden (not realistic threat model but could be interesting), pretty easy
- Show efficiency curves of ES vs Gradient-based (should probably def do this, track it while we go). Compare needed number of queries for each optimizer etc.
- Try more than the default target image, easy
- see if perturbations persist or get replaced with time once we stop attacking, easy

2 Actual paper starts here, put your stuff:

Problem Statement

Related Work

Mordvintsev et al. [1] introduced a growing Neural Cellular Automata (NCA) model which shows that small, shared-weight networks can grow images from a single center pixel. Additionally, by using sample pool training strategy and subjecting models to damage during training, the models become robust and show self-healing properties as the learned local update rules can regenerate damaged parts. Due to its robustness, NCA is being used to protect against adversarial attacks. For example, Xu et al. [2] inserted NCA between the middle layers of vision transformers in order to serve as an armor against adversarial attacks. In our project, we build on top of the growing NCA model. While the authors test their model's resilence to pixel deletions, we want to investigate how robust the model is against adversarial perturbations applied to the hidden cell states.

Existing research has explored how strong NCAs are against white-box attacks where attackers have full access to the model's internal weights. A study by Randazzo et al. [3] investigates such vulnerabilities of NCAs to white-box attacks and their results show that while growing cellular automata (CA) are more resistant to injections, they can still be reprogrammed using global state perturbations where mathematical transformations are applied to every cell's state at every step. Similarly, Cavuoti et al. [4] shows that a small number of adversarial cells that are injected into the image can drastically change its properties. Our approach also aims to explore the resilience of NCA, but instead of using gradient-descent approach we use a black-box evolutionary strategy that more closely resembles real-life attacks.

Covariance Matrix Adaptation Evolution Strategy (CMA-ES), introduced by Hansen [5], is particularly useful for black-box attacks. CMA-ES is an algorithm for non-linear, non-convex black-box optimization. It works by storing a population of candidates and iteratively shifting the population towards the patterns that cause the most significant damage to the output of the NCA. This way, new potential vulnerabilities can be revealed by evolving the perturbations over many generations. While evolutionary black-box attacks are commonly used for evaluating traditional image classifiers, they have not yet been used as an adversarial tool to disrupt CA. We will apply this general-purpose evolutionary method as the attacker in our project. By using CMA-ES, we test if evolutionary methods can find the same weaknesses as gradient methods.

Approach

- Train the standard NCA in Greydanus implementation, save weights
- Implement CMA-ES attacker
- Implement Random pertubations of same strength
- Probably hand in the first prototype comparing those
- Re-implement the attack from the Randazzo paper
- Also compare those
- Implement more from the optional list
- Write paper
- ???
- profit

Review Planning

Planning

To ensure steady progress and meet the project deadlines, we have structured our timeline into weekly milestones.

- **April 10 – April 17; Basis:** Set up the development environment and the Greydanus PyTorch NCA implementation. Successfully train a standard Growing NCA on a target image (e.g., the lizard), verify its self-organizing properties, and save the model weights.
- **April 17 – April 24; Prototyping:** Implement the attack surface logic (the 16×16 state-perturbation matrix). Implement the "Random Noise" baseline attacker. Integrate a basic CMA-ES loop to prove the end-to-end pipeline works (from perturbation generation $\rightarrow$ NCA rollout $\rightarrow$ L2 fitness evaluation).

  Milestone: Working Prototype comparing Random Noise by April 20.
- **April 24 – May 1; Gradient Baseline & Tuning:** Re-implement the white-box gradient-based attack from Randazzo et al. (or a standard Projected Gradient Descent on the perturbation matrix). Concurrently, tune the CMA-ES hyperparameters (population size, initial variance) for optimal targeted attack performance.
- **May 8 – May 22; Experiments & Data Collection:** Run the core comparative experiments across all three conditions (CMA-ES vs. Gradient vs. Random). Collect quantitative data, specifically tracking efficiency curves (number of queries/forward passes versus the resulting L2 distance from the targeted "broken" state).
- **May 22 – June 5; Continue Experiments & Paper Drafting:** Continue with list of Experiments. Draft the Results, Discussion, and Conclusion sections of the paper.

  Milestone: Project reaching maturity with a first draft by May 29.
- **June 5 – June 12; Finalizing & Minor Tests:** Run minor, final experiments. Format the final paper, review against the grading criteria, and submit.

References

[1] A. Mordvintsev, E. Randazzo, E. Niklasson, and M. Levin, "Growing neural cellular automata," *Distill*, vol. 5, no. 2, e23, 2020. [Online]. Available: https://distill.pub/2020/growing-ca doi: 10.23915/distill.00023.

[2] Y. Xu, T. Zhang, and S. Süsstrunk, "AdaNCA: Neural cellular automata as adaptors for more robust vision transformer," *Advances in Neural Information Processing Systems*, vol. 37, pp. 25709–25746, 2024.

[3] E. Randazzo, A. Mordvintsev, E. Niklasson, and M. Levin, "Adversarial reprogramming of neural cellular automata," *Distill*, vol. 6, no. 5, e00027-004, 2021. [Online]. Available: https://distill.pub/selforg/2021/adversarial doi: 10.23915/distill.00027.004.

[4] L. Cavuoti, F. Sacco, E. Randazzo, and M. Levin, "Adversarial takeover of neural cellular automata," in *Artificial Life Conference Proceedings 34*, 2022, p. 38. MIT Press.

[5] N. Hansen and A. Ostermeier, "Completely derandomized self-adaptation in evolution strategies," *Evolutionary Computation*, vol. 9, no. 2, pp. 159–195, 2001.